LAB SHEET

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SHEET - 3

Objectives

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

➢ **Commands:**

- • **TCL commands**: COMMIT, ROLLBACK, SAVEPOINT are used to control transactions.
- • **DCL commands**: GRANT and REVOKE are used to give or remove user privileges.
- • **GROUP BY** is used for aggregation.
- • **ORDER BY** is used for sorting.
- • **JOIN** is used to combine data from multiple tables.

- **DDL Commands**

1. Create TEACHER Table

```sql
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

2. Create SALES Table

```sql
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount DECIMAL(10,2),
    DOB DATE
);
```

3. Create DEPARTMENTS Table

```sql
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

4. Create DEPARTMENTS Table

```sql
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

- **DML Operations**

1. Insert 5 Records

```sql
INSERT INTO TEACHER VALUES
('Ravi',1,'2015-06-10','Mathematics','Delhi',50000),
('Anita',2,'2017-03-15','Commerce','Mumbai',45000),
('Karan',1,'2018-09-20','Mathematics','Delhi',55000),
('Neha',3,'2016-01-12','Science','Chandigarh',48000),
('Amit',2,'2019-11-05','Commerce','Mumbai',47000);
```

```sql
INSERT INTO SALES VALUES
(1,'Rohan','Delhi',20000,'1999-12-21'),
(2,'Priya','Mumbai',15000,'2000-05-10'),
(3,'Aman','Delhi',18000,'1998-12-15'),
(4,'Neha','Chandigarh',22000,'1997-08-25'),
(5,'Karan','Mumbai',25000,'1996-12-05');
```

```sql
INSERT INTO DEPARTMENTS VALUES
(1,'Mathematics','Delhi'),
(2,'Commerce','Mumbai'),
(3,'Science','Chandigarh');
```

```sql
INSERT INTO EMPLOYEES VALUES
(101,'Rohit',60000,1),
(102,'Priya',45000,2),
(103,'Aman',50000,1),
(104,'Neha',55000,3),
(105,'Karan',40000,2);
```

2. 25% Increment for Mathematics Department

```sql
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

3. 15% Increment for Commerce Department

```sql
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

- **From SALES Table**

1. Total Sales in Each Branch

```sql
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Output:

Result Grid | Filter Rows:

Branch	Total_Sales
Delhi	38000.00
Mumbai	40000.00
Chandigarh	22000.00

2. **Average Sales in Each Branch**

```sql
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Output:

Result Grid | Filter Rows:

Branch	Average_Sales
Delhi	19000.000000
Mumbai	20000.000000
Chandigarh	22000.000000

3. Salesname and DOB (Born in December)

```
SELECT Salesname, DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Output:

Result Grid | Filter Rows:

Salesname	DOB
Rohan	21-Dec-99
Aman	15-Dec-98
Karan	05-Dec-96

4. Salesname and DOB in Alphabetical Order of Month

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

Output:

Result Grid | Filter Rows:

Salesname	DOB
Priya	2000-05-10
Neha	1997-08-25
Rohan	1999-12-21
Aman	1998-12-15
Karan	1996-12-05

- **JOIN Operations**

1. Employee and Department Details (INNER JOIN)

Result Grid | Filter Rows: | Export: | Wrap

emp_id	emp_name	emp_salary	dept_name	dept_location
101	Rohit	60000.00	Mathematics	Delhi
103	Aman	50000.00	Mathematics	Delhi
102	Priya	45000.00	Commerce	Mumbai
105	Karan	40000.00	Commerce	Mumbai
104	Neha	55000.00	Science	Chandigarh

```
SELECT *
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Output:

Result Grid | Filter Rows:

emp_name	dept_name
Rohit	Mathematics
Priya	Commerce
Aman	Mathematics
Neha	Science
Karan	Commerce

2. LEFT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Output:

Result Grid | Filter Rows:

emp_name	dept_name
Rohit	Mathematics
Aman	Mathematics
Priya	Commerce
Karan	Commerce
Neha	Science

3. RIGHT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows:

emp_name	dept_name
Rohit	Mathematics
Aman	Mathematics
Priya	Commerce
Karan	Commerce
Neha	Science

4. Employees with Salary Greater than Average

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

Output:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
101	Rohit	60000.00	1
104	Neha	55000.00	3
NULL	NULL	NULL	NULL

- **TCL Commands**

1. ROLLBACK
2. COMMIT
3. SAVEPOINT

```
SAVEPOINT A;
```

```
ROLLBACK;
```

```
COMMIT;
```

- **DCL Commands**

1. Grant All Privileges

- ```sql
  CREATE USER IF NOT EXISTS 'dept_user'@'localhost' IDENTIFIED BY '1234';
  ```
- ```sql
  GRANT ALL PRIVILEGES ON DBMS_LAB3.EMPLOYEES TO 'dept_user'@'localhost';
  ```

2. Grant Selected Privileges

- ```sql
  GRANT SELECT, INSERT ON DBMS_LAB3.EMPLOYEES TO 'dept_user'@'localhost';
  ```

3. Revoke All Privileges

- ```sql
  REVOKE SELECT, INSERT ON DBMS_LAB3.EMPLOYEES FROM 'dept_user'@'localhost';
  ```

4. Revoke Selected Privileges

```sql
REVOKE ALL PRIVILEGES ON DBMS_LAB3.EMPLOYEES FROM 'dept_user'@'localhost';
```